From Spectrum to Verdict

What the DEV plugin computes, the physics underneath it, and which of the numbers deserve to be believed.

Why this document exists. The evaluation tab shows roughly twenty numbers. One of them decides the verdict; most of the others are historical, and at least one is inverted with respect to its name. This document writes down what each is, in formulas, so that reading a report does not require reading the source — and puts the physics beside the algebra, because the metric only makes sense with both.

Its companions. *Capture Fidelity* covers the instrument — how a webcam becomes a spectrum. *Light, Pigment and Solvent* covers the sample — what is in the jar and what it does to a photon. This one starts where those finish, at the absorbance curve $A(\lambda)$, and ends at the verdict.

How to read it. Chapter 1 stands alone and contains the claim. Chapter 3 is the physics, chapter 5 the metric that matters; if you read only two, read those. Chapters 6–7 are colour and caveats. Everything historical has been moved to the appendices.

A note on the worked numbers. Two data sets carry every example:

	set	oil	runs	provenance
green	`20270729C`	fresh green	6 re-seats of one fill	`SPEC_capture_quality.md` §16.11.3
brown	`20260731A`	brown	6 re-seats of one fill	§16.13, "series D"

Both are post-rebuild: same instrument, protocol and dilution recipe. They are the most directly comparable green/brown pair on record. Quoted values are the **mean of the six runs**.

CONTENTS

1. The claim

1.1 The winning metric

One number decides the verdict. It is the **Pigment Index** — internally `Pigment ratio · linear baseline`:

$$\text{Pigment Index} = \frac{B_{\text{Soret}}}{B_{\text{Q}}}$$

B_X = mean absorbance in band X, measured above a baseline fitted through two anchor windows (§5).

with B_{Soret} over **440–460 nm** and B_{Q} over **560–580 nm**, both read on the de-spiked absorbance after subtracting a straight line fitted through **520–540** and **600–630 nm**.

1.2 What it achieves

	green `20270729C`	brown `20260731A`
Pigment Index	**12.251 ± 0.354**	**9.303 ± 0.131**
at the shipped threshold **T = 10.6**	**+4.83 σ** above	**+9.88 σ** below
misclassification *(t-distribution)*	0.027 %	0.009 %

The gap is **2.947 = 31.7 % of the brown mean**, against a pooled σ of 0.267 — **Cohen's *d* = 11.04**.

The two classes do not overlap, on any run, in either set.

1.3 What Cohen's *d* means

The figure is used throughout this document, so it is worth defining once. A raw gap between two class means is uninformative on its own: 2.947 units is impressive only if the measurement's own run-to-run scatter is much smaller than that. **Cohen's *d*** divides one by the other:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}, \qquad s_{\text{pooled}} = \sqrt{\frac{s_1{}^2 + s_2{}^2}{2}}$$

the difference between the two class means, in units of their pooled standard deviation.

The subscripts 1 and 2 label the two groups being compared — here **1 = green, 2 = brown**:

symbol	meaning	value here
$\bar{x}_1$	the **mean** of group 1 — the average of green's six runs	12.251
$\bar{x}_2$	the **mean** of group 2 — the average of brown's six runs	9.303
s_1	the **standard deviation** of group 1 — how much green's runs scatter about their own mean	0.354
s_2	the standard deviation of group 2 — brown's scatter	0.131
s_{pooled}	the two scatters combined into one, as the root-mean-square of s_1 and s_2	0.267

So $d = (12.251 - 9.303) / 0.267 = 2.947 / 0.267 = 11.04$. Note that only the **standard deviations** enter, not the sample sizes: *d* describes how far apart the two *populations* look, not how confident we are about it. Confidence comes from n, and is handled separately in §5.8.

Read out loud, it is **"how many standard deviations apart the two groups are"**. It is dimensionless, so it does not care what units a metric carries, and it is therefore directly comparable **between rival metrics**. That is why every metric in this document is scored with it.

d	conventional reading	overlap of two normal distributions
0.2	small effect	almost complete
0.5	medium	heavy
0.8	**large**	substantial
3	very large	the distributions barely touch
11.04	**our Pigment Index**	**none observed**

Our *d* = 11.04 means the class means sit eleven pooled standard deviations apart — which is why no run of either set comes near the other's range.

⚠ Three reading notes. With n = 6 per class the *value* of *d* is loosely estimated, even though its sign and rough size are not in doubt; §5.8 gives the interval that actually matters. A **negative** *d* in this document means the ordering is **inverted** — brown reading higher than green. And *d* comes in more than one recipe: **Appendix B** gives both pooled-SD conventions, the small-sample correction, and which is used where.

The same idea outside statistics

The identical construction is standard in several other fields under other names — worth knowing, because a reader from any of them already understands the figure. The neutral, field-independent term for the whole family is the **standardised mean difference (SMD)**; Cohen's *d* is one recipe within it (Appendix B).

name	field	relation
d′ ("d-prime")	signal detection, psychophysics	the same quantity; the sensitivity of a detector
Fisher's discriminant ratio	pattern recognition	$J = d^2/2$ — here 60.9
Z-score, "sigma level"	process control, Six Sigma	distance from a *limit*, not between two groups
peak resolution R_s	chromatography	separation ÷ typical width — the same shape of idea

The Z-score row is already in use here without the label: the **"+4.83 σ above T"** and **"+9.88 σ below T"** of §1.2 are Z-scores — one group measured against a threshold, rather than two groups against each other.

1.4 Three properties that make the claim worth making

> **1. It is dilution-invariant *by construction*, not by luck.** Every step from the curve to the two band means is linear and homogeneous, so concentration and path length cancel exactly in the quotient. This is proved, not asserted (§5.6). It means the recipe can change without recalibrating the verdict.
>
> **2. It is a ratio of two chemical SPECIES, not a measure of "how much pigment".** What it tracks is intact protochlorophyll against its magnesium-free degradation product protopheophytin. Across our two classes the *total* Q-region absorbance is identical (0.2300 vs 0.2251) while the shape differs at d = 10.3 — nothing is missing, something has been **converted** (§3.3).
>
> **3. It survives the instrument.** Lamp brightness, camera gain, grating efficiency and exposure all scale the absorbance uniformly and divide straight out. A €30 webcam can carry it.

⇒ **Green-versus-brown discrimination works.** That is the claim, and §5.8 is where it is defended.

1.5 What the claim is *not*

⚠ **Precision is not correctness.** d = 11.04 says the instrument reliably distinguishes *these two bottles*. Whether **T = 10.6** is the right place to cut the population of real pumpkin oils is a separate, still-unvalidated question requiring reference oils with independent ground truth (`SPEC_capture_quality.md` §16.10.11a). A precise instrument reading a wrong threshold is confidently wrong.

⚠ **These are re-seat numbers.** Both sets are six re-seats of *one fill*, so they exclude sample preparation entirely. Fill-to-fill scatter remains unmeasured for brown (`SPEC_capability_proof.md` §11.4f B).

1.6 The chain, in one line

$$2\ captures \Rightarrow R(\lambda),\ S(\lambda) \Rightarrow A(\lambda) \Rightarrow \text{despike} \Rightarrow 4\ band\ means \Rightarrow the\ index$$

2. From two captures to an absorbance curve

2.1 The burst mean

Each capture is a burst of **150 frames** (`DevSpectralPlugin.FRAMES`). Frames whose overall brightness is an outlier are rejected, then each wavelength bin is averaged with a sigma-clipped mean:

$$R(\lambda) = \mathrm{mean}_{f \in F_{kept}}\, r_f(\lambda) \qquad S(\lambda) = \mathrm{mean}_{f \in F_{kept}}\, s_f(\lambda)$$

F_{kept} excludes frames rejected by the per-frame MAD test (SPEC_capture_quality.md §14.8 C1).

The reference is pure isopropanol in the same jar, measured in the same session. Dividing by it removes the lamp's own spectrum — the single most important idea in the instrument.

2.2 Transmittance, with a floor guard

$$T(\lambda) = \frac{S(\lambda)}{R(\lambda)} \qquad \text{for } R(\lambda) > f \cdot \max_\lambda R(\lambda), \qquad f = 6.31 \times 10^{-5}$$

Where the reference is at or below that fraction of its own peak the wavelength is **masked out entirely** rather than divided; below the floor S/R amplifies noise without bound. The constant looks odd because it lives in linear light — it is the historical "1 % of peak" expressed after gamma decoding, $0.01^{2.2}$ (`SPEC_capture_quality.md` §17).

2.3 Absorbance

$$A(\lambda) = -\log_{10} T(\lambda) = -\log_{10} \frac{S(\lambda)}{R(\lambda)}$$

defined only where T(λ) > 0; other wavelengths carry no value at all.

This is the **Beer–Lambert** quantity, $A = \varepsilon\, c\, l$ — linear in concentration, which is what makes ratios of it behave (*Light, Pigment and Solvent* §2.2).

2.4 De-spiking

Every metric except the "raw" twin rows is computed on the **de-spiked** curve — a moving median with a 7-point kernel:

$$A_d(\lambda_i) = \mathrm{median}\{A(\lambda_{i-3}), \ldots, A(\lambda_{i+3})\}$$

A median *rejects* isolated outliers where a smoothing filter would average them in. The grid spacing is 0.146 nm, so the kernel spans ≈1 nm — narrow enough to leave a 20–30 nm pigment band untouched, wide enough to kill single hot pixels.

⚠ **It does not remove the two named instrument artifacts.** Two narrow features near **473 nm** (FWHM 1.0 nm) and **607 nm** (FWHM 2.7 nm) are wider than the kernel and survive it. Both are **lamp emission lines** that fail to cancel in S/R (§5.9). The 473 nm one sits outside every measurement window and is harmless; the 607 nm one does **not** — it lies inside the 600–630 anchor.

3. The physics the metric rests on

This chapter is the short form. The full chemistry — miscibility, turbidity, the vessel — is in Light, Pigment and Solvent, whose §3 this compresses.

3.1 The pigment is protochlorophyll, not chlorophyll

Fruhwirth & Hermetter's review of the Styrian oil pumpkin identifies the oil's colourants explicitly: *"various tetrapyrrol-type compounds like **protochlorophyll (a and b)** and **protopheophytin (a and b)**, the latter being a protochlorophyll lacking the magnesium ion"* [1].

Protochlorophyll is chlorophyll's biosynthetic **precursor**. It lacks chlorophyll's ring-D reduction, so it is a **porphyrin** where chlorophyll is a **chlorin** — and that single bond moves the red absorption band by ~40 nm:

	Soret	red (Qy) band	fluorescence
chlorophyll *a (the textbook default — **not** ours)*	≈ 430 nm	≈ 662–665 nm	≈ 668–675 nm
protochlorophyll(ide) *a* *(ours)*	**≈ 432–440 nm**	**≈ 623–626 nm**	**≈ 630–636 nm**

Protochlorophyllide's Qy is measured at **623 nm** in 80 % acetone and **626 nm** in methanol [2], and the oil's own fluorescence maximum of **635 nm** [1] corroborates it — a fluorescing molecule emits ~10 nm to the red of the transition it absorbs on, putting the absorber at ~625 nm and nowhere near 662.

This matters because our capture window ends at 630 nm. The red band is therefore *at the edge of what we can see*, not beyond it — which is the whole reason the 600–630 window behaves as it does (§3.4).

3.2 Soret and Q — where the bands come from

All porphyrin-type spectra share one two-part shape, explained by **Gouterman's four-orbital model** [3]: the visible spectrum is governed by four frontier orbitals, two nearly-degenerate occupied and two nearly-degenerate empty.

band	transition	character
Soret (also **B**)	$S_0 \rightarrow S_2$	**very strong**, blue, ~430–440 nm
Q	$S_0 \rightarrow S_1$	**weak** — 1/5 to 1/10 of the Soret — yellow-green to red

In a metalloporphyrin the magnesium sits on a four-fold symmetry axis (D_{4h}). The two Q transitions are then **degenerate** — equal energy, polarised at right angles in the plane of the ring — and what appears is one origin band **Q(0,0)** (also α) plus a vibronic satellite **Q(1,0)** (β).

> **"Degenerate" carries no sense of decay.** It means two states happen to share an energy. Symmetry is what enforces the sharing, so lowering the symmetry **lifts** the degeneracy and the states separate.

3.3 What roasting does — and why it rearranges rather than removes

Heat and acid **strip the central magnesium out of the ring**, replacing it with two protons. The product is **protopheophytin** (*pheo-*, Greek *phaiós*, "dusky"). It is not only roasting: in pumpkin seeds protopheophytins accumulate as a **storage** degradation product, reported at **1.1–35.5 %** of the protochlorophylls [4].

Spectroscopically the two protons sit on **one** axis, so the symmetry drops $\mathbf{D_{4h}} \rightarrow \mathbf{D_{2h}}$, the degeneracy is lifted, and the two Q transitions separate into distinct **Qx** and **Qy** bands. Each keeps its own vibronic satellite, so a metal-free ring shows **four** Q bands, numbered **I–IV from the longest wavelength**: I = Qy(0,0), II = Qy(1,0), III = Qx(0,0), IV = Qx(1,0).

⭐ **And the redistribution has a known direction.** Free-base porphyrin spectra are classified into four types by Q-band intensity ordering — *etio* (IV > III > II > I), *rhodo* (III > IV > II > I), *oxo-rhodo* and *phyllo* [5]. **Band I, the longest-wavelength band, is the weakest in every one of them.** So a pigment whose long-λ band *dominates* while metallated finds it demoted to weakest-of-four once the magnesium goes. Intensity moves toward the blue. *(Protopheophytin carries the ring-E carbonyl, a "rhodofying" group, so* rhodo *is its expected type — band I still weakest.)*

⇒ **The verdict is a ratio of two chemical species**, intact against degraded — not a measure of how much pigment is present.

3.4 ⭐ Why this appears as a change of SLOPE

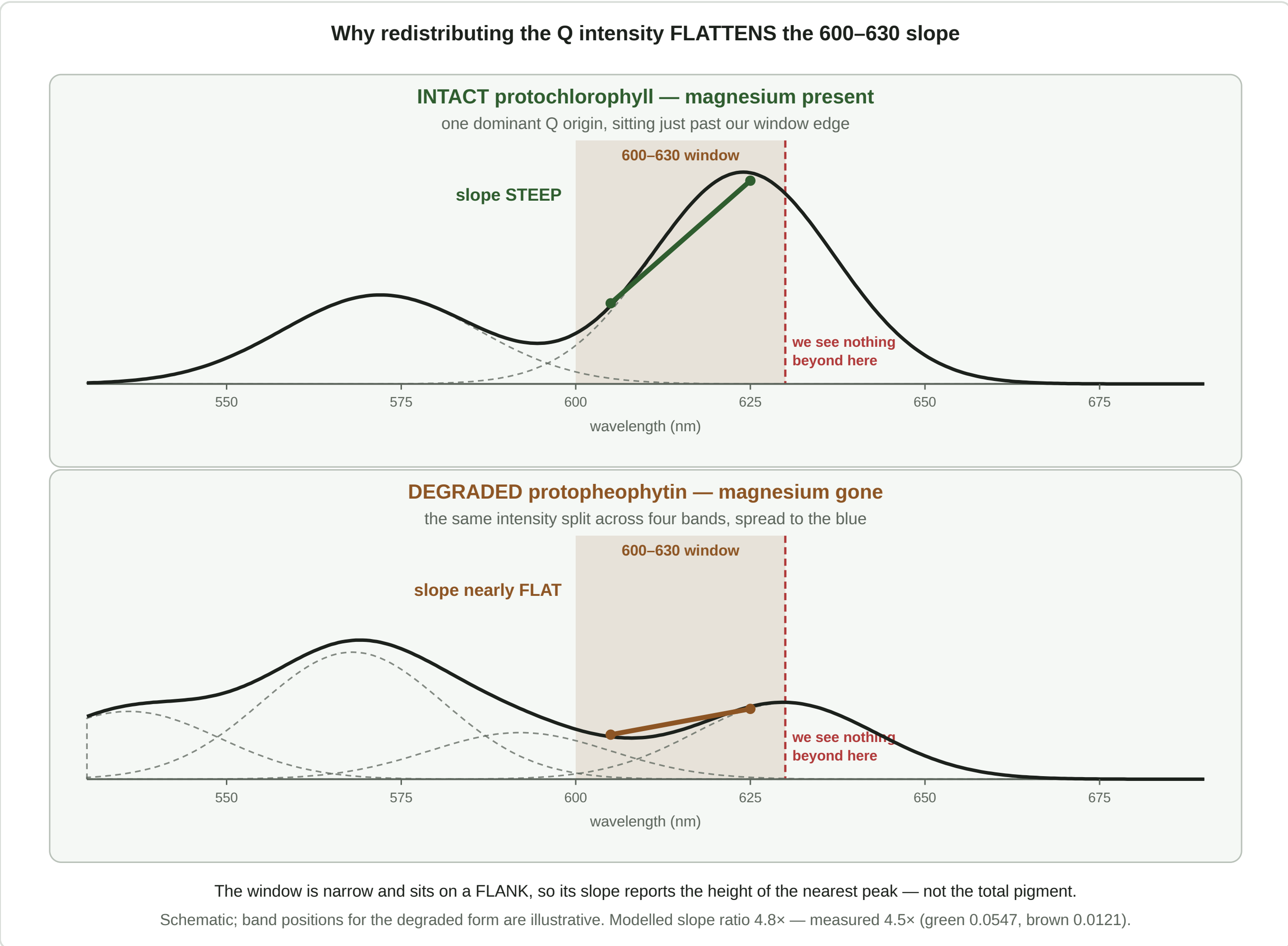

Figure 1 — the mechanism. **Top:** with the magnesium intact, one dominant Q origin sits just past our capture limit and its rising flank sweeps through the 600–630 window, so absorbance climbs steeply across it. **Bottom:** once the magnesium is gone that intensity is spread across weaker bands further blue; nothing tall is left near 630, the window lies on a shallow shoulder, and the slope nearly vanishes. Total area is similar — only its distribution differs. Schematic, but the modelled slope ratio (4.8×) matches the measured one (4.5×).

Our 600–630 window is **narrow and sits on a flank, not on a peak**. The slope across such a window reports **the height of the nearest peak**, not the total pigment. A tall band whose edge crosses the window gives a steep rise; move that intensity 30–50 nm blue and split it, and the window is left on flat ground even though just as much light is absorbed overall.

That makes the far window an unusually **specific** probe: it responds to *whether the intact, symmetric pigment is still there*, and is comparatively blind to how much total pigment the oil contains.

3.5 The far window is a measuring band — measured, not assumed

The 600–630 window was introduced as an "oil-quiet" anchor. It is not quiet. Across 37 runs, six fills and two sessions under an identical lamp, the *rise* $A(620\!-\!630) - A(600\!-\!610)$ is **0.0535 for green against 0.0159 for brown — a 5.1 σ separation** (`SPEC_capture_quality.md` §16.12.12). The reference sits at 35–39 DN in both classes, so the lamp is in the same state: an instrument artifact cannot know which oil is in the jar.

And it is load-bearing. Sweeping the far window's right edge inward, Cohen's *d* falls **2.88** → **0.94** and the classes overlap outright by 600–610 nm (§16.12.13). The contamination *carries* the discrimination.

> ⛔ **Changing these windows is a metric redesign, not a tidy-up**, and it is gated on post-rebuild data (`SPEC_capture_quality.md` §16.12.16).

3.6 The difference is speciation, not concentration — and the test has no free parameters

Under simple Beer–Lambert with one absorbing species, brown = *k* × green: the curves differ by **one scale factor**, so any ratio of two features taken *inside* the Q region is class-independent. A class difference in such a ratio refutes pure scaling outright.

	green	brown	*d*
`A_Q` 560–580	0.2300 ± 0.0173	0.2251 ± 0.0199	**0.26** — equal
Q amplitude (572 − 550)	0.1275	**0.1495**	**−3.16** — brown HIGHER
rise ÷ Q-amplitude	**0.4274 ± 0.039**	**0.0800 ± 0.028**	**10.26**

Pure scaling predicts *d* = 0 on the last row. Measured *d* = 10.26, a factor of 5.3. Brown is not pigment-poor — `A_Q` is equal and its 572 feature is *stronger*. Normalising each class by its own Q amplitude shows where the intensity went: **out of 615–630 (−0.193) and into 580–615 (+0.07 to +0.11)**, exactly the direction §3.3's band-I rule predicts. *(`diagnostics/qband_shape.py`; `SPEC_capture_quality.md` §16.13.9.)*

3.7 ⚠ Why we cannot simply look at the two-versus-four bands

The natural objection is that a better spectrometer would settle all of this. **It would not:**

grid spacing	0.146 nm/bin, 1305 bins over 440–630
narrowest feature resolved *(473 nm lamp line)*	**FWHM 1.0 nm**
second lamp line *(607 nm)*	**FWHM 2.7 nm**
Q bands to be separated	**20–30 nm wide**

The rig **out-resolves the target by 10–20×**. Four other things stand in the way: **window truncation** at 630 nm (the dominant limit); the two species **always coexisting**, so no pure spectrum of either is ever presented;

20–30 nm intrinsic linewidths merging the four free-base bands into shoulders; and the **turbidity pedestal** suppressing contrast.

⇒ **The remedy is a wider window, not a finer instrument** — an argument that redirects an entire class of "buy better hardware" thinking toward a much cheaper change.

4. The windows and the band means

Everything downstream is built from four wavelength windows — two pigment bands and two baseline anchors.

symbol	window	what it sits on	source
A_{Soret}	**440–460 nm**	red flank of the **Soret (B)** band, ~432 nm	`PB_SORET_BAND`
A_{Q}	**560–580 nm**	a band in the **Q region** — assignment open, §7.4	`PB_Q_BAND`
A_{near}	**520–540 nm**	between bands — the quieter anchor	`PB_BASELINE_WINDOWS[0]`
A_{far}	**600–630 nm**	the approach to **Qy(0,0)** — a *measuring* band (§3.5)	`PB_BASELINE_WINDOWS[1]`

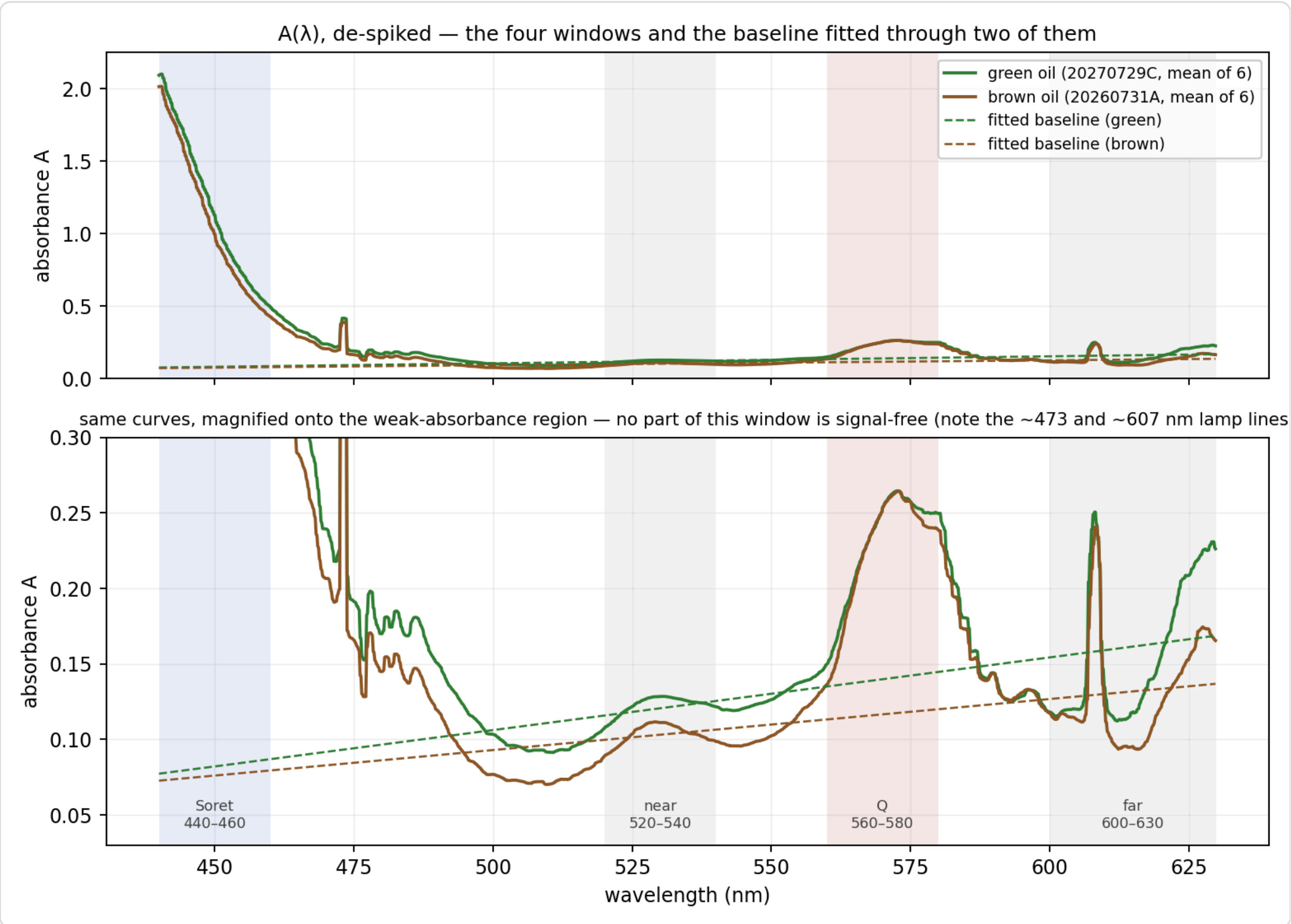

Figure 2 — the de-spiked absorbance of both classes, with the four windows shaded and each curve's own fitted baseline drawn dashed. The lower panel magnifies the weak-absorbance region — **no part of this window is signal-free** (§3.5). Note how little separates the two curves inside the Q window, and how much separates them inside the far window.

4.1 Windows and band means — the notation used from here on

A **window** is a wavelength interval. Write W_X for the *set of measured grid points* that fall inside window X:

$$W_X = \{\lambda : lo_X \le \lambda \le hi_X\}$$

read: "all wavelengths λ between the window's lower and upper edge". $|W_X|$ is how many there are.

So W_{Soret} is the 138 grid points between 440 and 460 nm, W_{far} the 212 points between 600 and 630 nm, and so on. The **band mean** is then simply their average:

$$A_X = \frac{1}{|W_X|} \Sigma_{\lambda \in W_X} A_d(\lambda)$$

the sum of the de-spiked absorbance over the window's points, divided by their number.

A plain unweighted mean of every surviving grid point — **not** an integral. For two 20 nm bands the choice makes no difference to the ratio, and a mean keeps the same unit as the other band readings, where an integral would inject a bandwidth factor into the unequal-width comparisons (`SPEC_pumpkin_peak_ratio_eval.md` §9).

4.2 Geometry and measured values

window	points	centroid $\bar{\lambda}$	green	brown	CV green	CV brown
Soret 440–460	138	449.98 nm	1.1864	1.0855	4.04 %	3.58 %
Q 560–580	137	570.02 nm	0.2300	0.2251	7.52 %	8.86 %
near 520–540	135	529.93 nm	0.1231	0.1038	8.89 %	13.31 %
far 600–630	212	615.06 nm	0.1591	0.1311	13.97 %	15.19 %

The centroids matter in §5.5. **Read the A_Q row**: 0.2300 against 0.2251, a 2 % difference against 7–9 % scatter. On its own the Q band carries no usable class information (d = 0.26).

5. ⭐ The Pigment Index — the metric that works

5.1 What problem the baseline solves

Re-seating the jar tilts the beam slightly, and that enters the absorbance curve as **an offset and a slope together** — the whole curve lifts *and* rotates. A constant subtraction removes the offset; standard normal variate removes offset and scale; **neither removes a slope.** A straight line fitted through two separated anchor windows removes both (`SPEC_capture_quality.md` §16.10.2).

5.2 The definition

Take every grid point inside **either anchor window** — W_{near} (520–540 nm, 135 points) and W_{far} (600–630 nm, 212 points), in the notation of §4.1 — and fit **one straight line** through all 347 of them at once.

The line is $A = m\lambda + c$, with $\boldsymbol{m}$ **its slope** in absorbance per nanometre and $\boldsymbol{c}$ **its intercept** — the value the line would take at $\lambda = 0$ nm. The intercept has no physical meaning on its own; it is simply the second number needed to pin a line down once the slope is chosen. What matters is the line's *value across our window*, which §5.3 tabulates.

The fit is **weighted least squares**, with each of the two windows carrying **equal total weight** regardless of how many points it contains. *(What that means and why squares: Appendix A.)*

Subtract the fitted line from the **whole** curve, re-read the two pigment bands on the corrected curve, and divide:

$$B(\lambda) = A_{\mathrm{d}}(\lambda) - (m\lambda + c)$$

$$B_{\mathrm{Soret}} = \frac{1}{|W_{\mathrm{Soret}}|} \Sigma_{\lambda \in W_{\mathrm{Soret}}} B(\lambda) \qquad B_{\mathrm{Q}} = \frac{1}{|W_{\mathrm{Q}}|} \Sigma_{\lambda \in W_{\mathrm{Q}}} B(\lambda)$$

$$\text{Pigment Index} = \frac{B_{\mathrm{Soret}}}{\max(B_{\mathrm{Q}},\ \varepsilon)}, \qquad \varepsilon = 10^{-3}$$

the epsilon is a division guard, never reached in practice.

5.3 What the correction does to each band

Measured fits, mean over each set:

	slope m	intercept c	baseline under Soret	baseline under Q
green	+4.808 × 10^{-4} A/nm	−0.1342 A	0.0822	**0.1399**
brown	+3.376 × 10^{-4} A/nm	−0.0758 A	0.0761	**0.1166**

$$B_{\mathrm{Soret}} = 1.1864 - 0.0822 = 1.1043 \qquad B_{\mathrm{Q}} = 0.2300 - 0.1399 = 0.0901 \qquad (green)$$

	$A_{Soret} \rightarrow B_{Soret}$	$A_Q \rightarrow B_Q$
green	1.1864 → 1.1043 *(−7 %)*	0.2300 → **0.0901** *(−61 %)*
brown	1.0855 → 1.0093 *(−7 %)*	0.2251 → **0.1085** *(−52 %)*

The Soret barely notices; the Q band loses more than half its value. That asymmetry follows from geometry — the baseline is a small fraction of a large absorbance and a large fraction of a small one — and it is the whole story of §5.4.

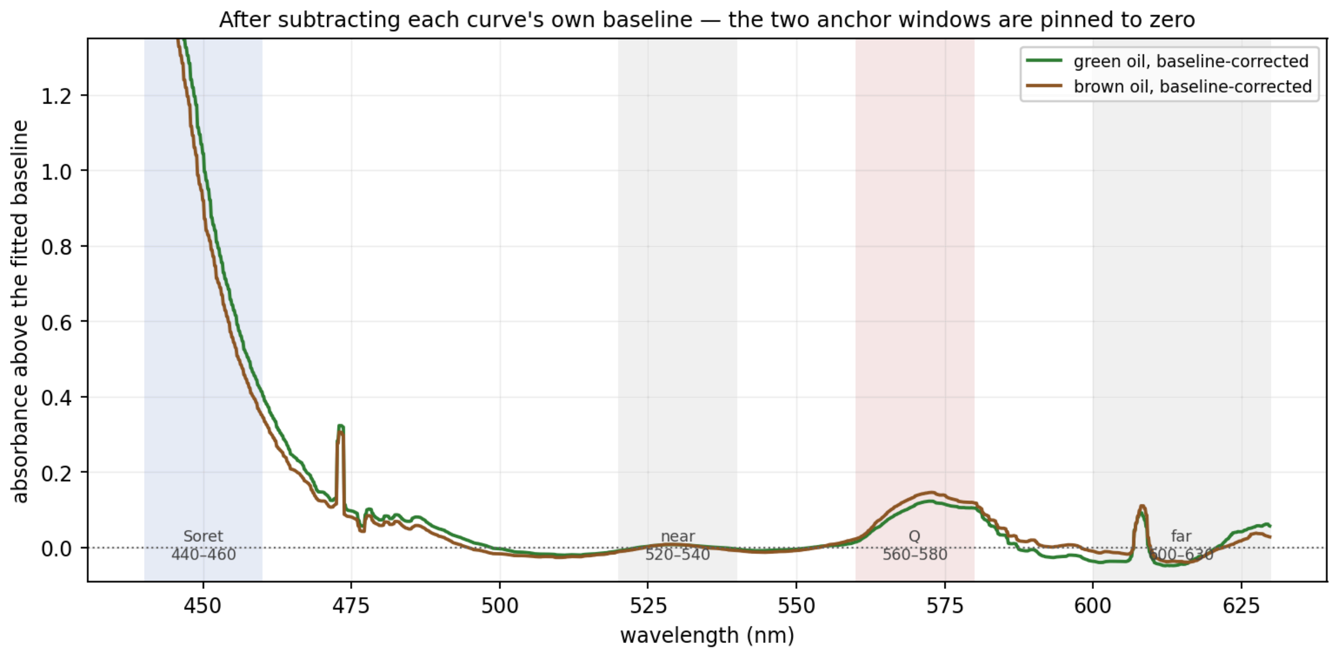

Figure 3 — the same two curves after each has had its own baseline subtracted. The two anchor windows are pinned to zero by construction; everything else is now measured relative to them.

5.4 ⭐ Why it discriminates — the denominator inverts

	green	brown	green ÷ brown	*d*	separation
A_Q *(before)*	0.2300	0.2251	1.02	0.26	overlap
B_Q *(after)*	**0.0901**	**0.1085**	**0.83**	**−6.48**	**clean, inverted**
B_{Soret}	1.1043	1.0093	1.09	2.44	clean
Pigment Index	**12.251**	**9.303**	**1.32**	**11.04**	**clean**

Before correction the two classes' Q bands are indistinguishable. After correction the ordering **reverses** and becomes decisive: brown's corrected Q band is 20 % *higher*, at *d* = −6.48 with no overlap between the two sets of six runs.

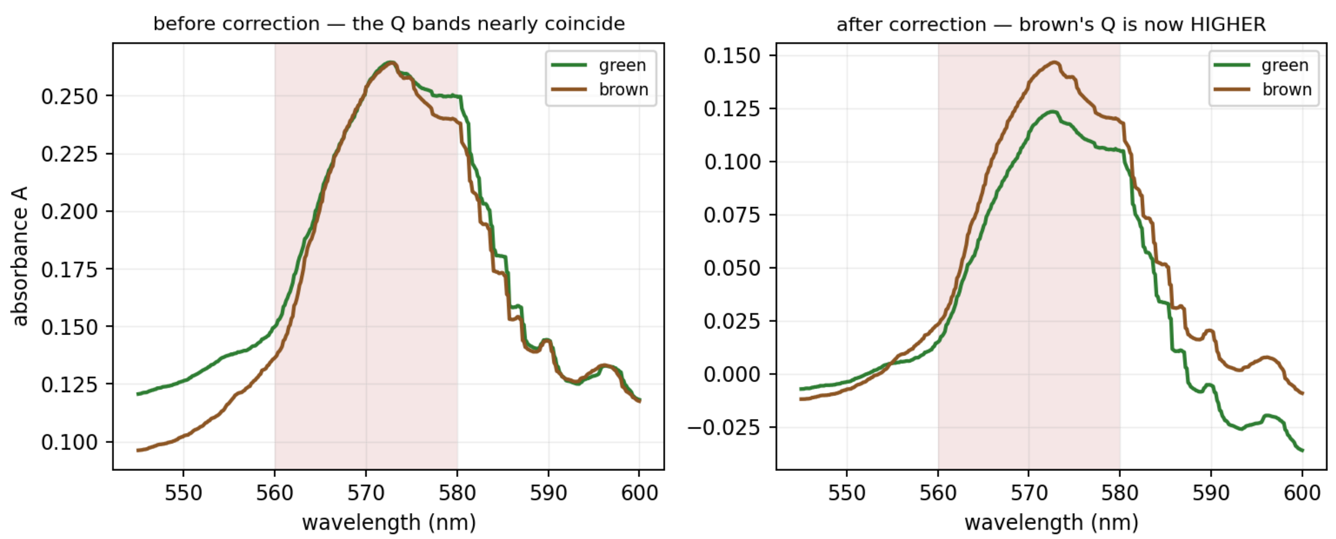

Figure 4 — the Q band before and after correction. Left: the curves lie on top of each other inside the shaded window. Right: after each curve's own baseline is removed, brown stands clearly above green. Nothing about the oils changed between the panels — only what the band is measured *above*.

Why. The baseline under the Q band differs by class — **0.1399** green against **0.1166** brown — so green has *more* subtracted. And green's baseline sits higher because A_{far} is 0.1591 against 0.1311: 600–630 carries real intact pigment (§3.5). The causal chain:

$$\textit{intact pigment raises } A_{\text{far}} \Rightarrow \textit{raises the fitted baseline} \Rightarrow \textit{lowers } B_{\text{Q}} \Rightarrow \textit{raises the index}$$

The two effects multiply:

$$\frac{\text{index}_{\text{green}}}{\text{index}_{\text{brown}}} = \frac{B_{\text{Soret,green}}}{B_{\text{Soret,brown}}} \times \frac{B_{\text{Q,brown}}}{B_{\text{Q,green}}} = 1.094 \times 1.204 = 1.317$$

The numerator contributes ×1.09 and the **denominator ×1.20** — the inverted Q band is the larger term. A *d* = 2.44 numerator and a *d* = −6.48 denominator compose into *d* = 11.04.

This is why the uncorrected ratio fails (Appendix D.1): it divides a signal-bearing Soret band by a Q band that carries nothing. Only after the baseline is removed does the denominator become a discriminator — and then the dominant one.

5.5 The three-region identity

Because the baseline is a straight line read at the band centroids, the metric can be written without mentioning a baseline at all. With t_{X} the position of band X between the anchor centroids:

$$t_{\text{X}} = \frac{\bar{\lambda}_{\text{X}} - \bar{\lambda}_{\text{near}}}{\bar{\lambda}_{\text{far}} - \bar{\lambda}_{\text{near}}} \qquad t_{\text{Soret}} = -0.939 \qquad t_{\text{Q}} = +0.471$$

$$B_X \approx A_X - [(1 - t_X)\,A_{\text{near}} + t_X\,A_{\text{far}}]$$

$$\text{Pigment Index} \approx \frac{A_{\text{Soret}} - 1.939\,A_{\text{near}} + 0.939\,A_{\text{far}}}{A_Q - 0.529\,A_{\text{near}} - 0.471\,A_{\text{far}}}$$

Verified against the shipped code: max error 0.52 % (green), 0.21 % (brown).

Read the signs. A_{far} enters the numerator **positively** and the denominator **negatively** — both *raise* the index. This is not "Soret over Q with a correction"; it is a **three-region measurement** in which the red window participates twice, with the largest coefficient of any numerator term save the Soret band itself.

> **Why the identity is approximate.** The real fit is a least-squares line through ~347 points and is **steeper** than the naive two-centroid line (+13.8 % green, +5.4 % brown), because both anchor windows have internal slope. It survives as a good approximation for a structural reason: the windows' weighted centroid sits at **572.5 nm** and the Q band's centroid at **570.0 nm** — 2.5 nm away. The extra steepness pivots almost exactly *on* the Q band, leaving B_Q nearly untouched while moving B_{Soret}, 122 nm away, by ~0.6 %. The two shifts partly cancel in the quotient.

5.6 ⭐ Dilution invariance — the proof

Model the curve as pigment plus a scattering pedestal that does **not** track concentration:

$$A(\lambda) = \varepsilon(\lambda)\,c\,l + P(\lambda)$$

ε extinction, c concentration, l path length, P the turbidity pedestal.

Take $P = 0$ for a moment. Every step from the curve to the two corrected band means — the band mean, the least-squares fit, the subtraction — is **linear and homogeneous** in the data. So scaling the input scales the output identically:

$$B_X = c\,l\,(\bar{\varepsilon}_X - \ell_\varepsilon(\bar{\lambda}_X)) \equiv c\,l\,e_X$$

$$\text{Pigment Index} = \frac{c\,l\,e_{\text{Soret}}}{c\,l\,e_Q} = \frac{e_{\text{Soret}}}{e_Q}$$

c and l are gone. A degree-1 homogeneous functional over a degree-1 homogeneous functional is degree-0. **This is invariance by construction, not by approximation.**

> **One step is exact and is easily mistaken for an approximation.** $B_X = A_X - L(\bar{\lambda}_X)$ holds *exactly*, because the mean of a straight line over a window equals that line at the window's centroid. The only approximation in §5.5 was the slope.

The proof never mentions chlorophyll. It uses only that a component scales with c — so §3.5's far-anchor pigment contamination is **harmless to invariance**. It changes what the metric *means* chemically; it does not change whether it is invariant.

5.7 What breaks invariance — pedestal curvature, and nothing else

Restore P. The fit is linear, so it splits:

$$B_X = c\,l\,e_X + r_X, \qquad r_X = \overline{P}_X - \ell_P(\overline{\lambda}_X)$$

r_X = the pedestal's departure from its OWN best-fit line, at band X.

pedestal	invariance
$P = 0$	exact
P exactly **linear** in λ	**also exact** — the fit removes it completely, $r \equiv 0$
P curved	broken, in proportion to r

So invariance fails only to the extent that turbidity is non-linear across 440–630 nm. Scattering goes roughly as λ^{-n} and a line approximates a curve; that curvature is the *entire* residual error.

The sensitivity, and the denominator is hit ten times harder again:

$$\frac{d\ \ln\ (\text{index})}{d\ \ln\ c} \approx \frac{r_Q}{B_Q} - \frac{r_{Soret}}{B_{Soret}} \approx \frac{r_Q}{B_Q}$$

The second term drops because B_{Soret}/B_Q is **12.3** (green) and **9.3** (brown).

And the error is concentration-dependent, since r_Q is fixed while B_Q grows with c:

$$\text{sensitivity} \approx \frac{r_Q}{c\,l\,e_Q} \propto \frac{1}{c}$$

Two consequences: a log–log plot of the index against concentration must be **curved**, flattening toward high c (a constant slope would falsify the pedestal explanation); and the total error is **U-shaped**, because curvature degrades invariance at *low c* while Soret stray-light compression degrades it at *high c* (`SPEC_capture_quality.md` §16.11.8). An optimal working concentration exists between them.

Against the record: the baseline **halves** the dilution error (5.49 % → 2.75 %); two dilutions 17 % apart in A_Q give means **1.9 %** apart, t = 1.14, p = 0.28. Taken at face value that implies $|r_Q| \leq sssim\ 0.01$ A — an **upper bound**, not a measurement. Full derivation: `SPEC_capture_quality.md` §16.14.

5.8 Performance

	value
green mean / σ *(n = 6)*	12.251 / 0.354
brown mean / σ *(n = 6)*	**9.303 / 0.131**

	value
gap	**2.947 = 31.7 %** of the brown mean
pooled σ	0.267
Cohen's *d*	**11.04**
green margin to T = 10.6	**+4.83 σ** → false-BROWN **0.027 %**
brown margin to T = 10.6	**+9.88 σ** → false-GREEN **0.009 %**

How well is σ known? With n = 6 the point estimate is not what to plan on. A χ^2 interval on brown's σ gives [0.082, **0.322**] — and **even at the upper bound brown clears the threshold by 4.03 σ** (false-GREEN 0.50 %). The conclusion does not depend on six points estimating σ well.

The brown mean also survived a rig rebuild and a different oil: archived `20260727C` read 9.361 on the old rig from six *fills*; series D reads **9.303**, a difference of **−0.62 %.**

(t-distribution throughout, per `SPEC_capture_quality.md` §16.10.11a — the error is heavy-tailed, so the Gaussian is optimistic exactly where it matters.)

5.9 ⚠ The 607 nm artifact sits inside the far anchor

Bridging it by interpolation across 604–612 nm and recomputing:

	A_{far} as measured	artifact bridged	contribution
green	0.1591	0.1484	**+7.2 %**
brown	0.1311	0.1209	**+8.4 %**

Its effect on the index is a **near-common-mode shift**: −5.9 % (green) and −4.9 % (brown) when removed, with *d* going 11.04 → 11.65 — slightly *better* without it. So the artifact is not creating the discrimination, but it **is** baked into the absolute scale on which **T = 10.6 was calibrated**.

What the artifact actually is

Both named artifacts are **lamp emission lines**, present in the reference spectrum itself:

	position in the reference	height above the local continuum
473 nm feature	~475 nm	**+48 %**
607 nm feature	~606 nm	**+20 %**

A sharp line in the lamp should *cancel* in $T = S/R$, since it appears in both captures. That it does **not** cancel — it survives into the absorbance as a spike — means the two captures do not see it identically. The project's term for this is **registration**: the reference and the sample are two separate jar insertions, and anything that displaces the beam between them (the jar seating, §5.1) shifts where a given wavelength lands on the sensor. A sharp line offset by a fraction of its own width turns into a derivative-shaped spike when the two spectra are divided. The artifact's 2.7 nm width is consistent with a shift of that order.

⚠ **This attribution is the project's, and I could not confirm it cleanly.** Estimating the line's position separately in reference and sample gives an apparent offset, but the estimate is confounded: the oil's own absorbance changes across the window, which tilts the local continuum and drags any centroid measure with it. **The lamp-line origin is verified; the registration mechanism is plausible and untested.** A competing explanation — detector nonlinearity or in-instrument stray light at a bright, high-contrast line — has not been excluded.

⇒ **What would move the threshold** is therefore a change to **the lamp's line spectrum** (a different lamp, or an ageing one) or to **the reference↔sample alignment** (calibration, jar seating, optics). A different *camera* is at most a second-order influence, through how the sensor grid samples the line — an earlier draft of this document overstated it.

6. Colour

The evaluation tab carries ten colour chips. They are a presentation feature, and this chapter exists partly to say why they are **not** a verdict input.

6.1 How a spectrum becomes a colour

$$\mathrm{SPD} \Rightarrow \mathrm{CIE}\ XYZ \Rightarrow xy \Rightarrow \mathrm{RGB} \Rightarrow \mathrm{HSL} \Rightarrow hue$$

The spectrum is rebinned to 380–780 nm at 1 nm, integrated against the **CIE 1931 2° colour-matching functions** under **D65**, reduced to chromaticity xy, converted to RGB and read as HSL (`SPEC_spectrum_processing.md` §4). **Luminance is discarded at the $XYZ \rightarrow xy$ step**, so every colour reported is chromaticity-derived.

Two converters are used deliberately: absorbance-derived chips use **sRGB** (full gamut), transmission-derived chips use the tuned **rgbxy** module (verdict-compatible, so the pumpkin hue thresholds are untouched). Absorbance values are sanitised first — non-finite → 0, **negatives → 0** (absorbance goes negative where $T > 1$, and negatives corrupt the CIE integral) — and capped by a **relative** ceiling at twice the 99th-percentile value, so a $T \rightarrow 0$ spike cannot dominate.

6.2 The three families

family	source	behaviour under dilution
Intrinsic	absorbance	**invariant** — $A \rightarrow kA$ leaves chromaticity unchanged
Intrinsic-perceived	absorbance, complemented	invariant
Perceived	transmission	**dilution-dependent** — $T \rightarrow T^k$

Intrinsic-perceived is the *colorimetric* complement: reflect the absorbed chromaticity through the D65 white point, $2 \cdot white - absorbed$ — the "other half of the light". This lands ~4° from the true perceived hue, against ~34° for a naive +180° HSL flip.

Each family is shown at its measured saturation and lightness plus a **hue-normalised** twin at fixed S = 38 %, L = 34 %, so that only hue varies between oils. A guard greys any chip whose **chroma** $(1 - |2L - 1|) \cdot S$ falls below 8 % — near-white and near-black have no meaningful hue, and HSL saturation alone would report a false one.

6.3 ⛔ Measured: colour does not discriminate these oils

chip	green *(n = 12)*	brown *(n = 6)*
Intrinsic	H 298–300°	H 298–300°
Intrinsic-perceived	H 67–69°	H 68–69°
Perceived	H 70°	H 70–71°
hue-normalised variants	H 300° · S 38 % · L 34 %	**identical**

This **confirms** `SPEC_capture_quality.md` §16.10.15 ("colour channels do NOT discriminate this oil pair") on post-rebuild data, and extends it from raw channels to the full HSL retrieval path.

Why it fails is instructive. Colour is a *broadband integral* — the CMFs weight the whole visible range — so the narrow, structured differences that carry the class signal (§3.6: a redistribution inside a 50 nm slice of the Q region) are averaged away against the enormous Soret absorbance that both oils share. The metric wins precisely because it looks at **narrow windows**; colour loses for the same reason.

⇒ The chips are worth showing and must never be thresholded.

7. What these numbers do not mean

7.1 A band mean is not dilution-invariant

A_{Soret} separates our two sets at *d* = 2.31 and B_{Soret} at *d* = 2.44, with no overlap. **Do not use them.** The two sets happen to share a dilution recipe, so the comparison is valid only *within* that accident. Only ratios divide the common factor out — which is why the UI marks ratios in bold as decision metrics and shows the band means without thresholds.

7.2 A ratio is invariant only if BOTH sides are corrected the same way

> **A ratio is dilution-invariant if and only if its numerator and denominator are both pedestal-free and corrected the same way.**

The legacy $G = D_{\mathrm{Q}}/A_{\mathrm{clarity}}$ and $G' = D_{\mathrm{Q}}/A_{\mathrm{blue}}$ apply the correction to **one side only** — a locally baseline-corrected numerator over an uncorrected denominator. They are not merely weak discriminators; they are *structurally* incapable of dilution invariance (`SPEC_capture_quality.md` §16.14.3).

7.3 The two gauges are not on the same scale

The tab shows two gauges: the first driven by the uncorrected Soret/Q (threshold 4.4), the second by the Pigment Index (threshold 10.3 in the gauge, T = 10.6 in the decision table). **Only the verdicts are comparable — never the numbers.** They are ratios of different quantities.

7.4 ⚠ We do not know what the 560–580 band is

Two candidates, and they are not equivalent: the **Q(1,0)** vibronic satellite of the intact pigment, or a **Qx** band of protopheophytin. Since 560–580 is the index's *denominator*, the difference is not academic — if it is the degradation product, the metric is a literal *intact* ÷ *degraded* ratio.

Evidence favours the second (§3.6: `A_Q` equal across classes while the 572 feature is *stronger* in brown), but **no source we hold assigns this band**, and the comparison is between two bottles rather than one oil before and after demetallation. Open: `SPEC_pumpkin_peak_ratio_eval.md` §15.5.

7.5 Precision is not correctness, and these are re-seat numbers

Both restated from §1.4 because they are the two most commonly dropped caveats. **T = 10.6 is unvalidated**; and σ_fill — the scatter a real single measurement is subject to — is unmeasured for brown.

8. Reference sheet

Green = `20270729C`, brown = `20260731A`, both the mean of six runs.

shown as	symbol	formula	green	brown	*d*
Soret · 440–460	A_{Soret}	mean A_{d} over 440–460	1.186	1.086	2.31
Q · 560–580	A_{Q}	mean A_{d} over 560–580	0.230	0.225	0.26
Clarity · 510–540	A_{clarity}	mean A_{d} over 510–540	0.114	0.095	1.62
Pigment ratio	—	$A_{\mathrm{Soret}} / A_{\mathrm{Q}}$	5.172	4.842	1.18
Pigment ratio · clarity	—	$A_{\mathrm{Soret}} / A_{\mathrm{clarity}}$	10.409	11.572	−1.08
Soret · linear baseline	B_{Soret}	mean B over 440–460	1.104	1.009	2.44
Q · linear baseline	B_{Q}	mean B over 560–580	0.090	0.109	−6.48
Pigment ratio · linear baseline	**Pigment Index**	$B_{\mathrm{Soret}} / B_{\mathrm{Q}}$	**12.251**	**9.303**	**11.04**
Greenness G	G	$D_{\mathrm{Q}} / A_{\mathrm{clarity}}$	—	—	−1.99
Pigment D_Q	D_{Q}	peak above a local line	—	—	−2.48
Soret A_blue	A_{blue}	reference-gated 450–490	—	—	1.18
Pigment ratio · legacy	—	$A_{\mathrm{blue}} / A_{\mathrm{clarity}}$	—	—	0.11
G' (alt.)	G'	$D_{\mathrm{Q}} / A_{\mathrm{blue}}$	—	—	−5.33

Constants

constant	value	where
capture window	440–630 nm	`WAVELENGTH_MIN_NM` / `MAX`
frames per capture	150	`FRAMES`
de-spike kernel	7 points ≈ 1 nm	`MedianFilterOp`
reference floor	6.31×10^{-5} of peak	`TransmissionLogicModule`
division guard ε	10^{-3}	`DevSpectralPlugin`
hue-normalised S / L	38 % / 34 %	`DevSpectralPlugin`
achromatic chroma guard	8 %	`EvaluationColorUtil`
verdict threshold T	10.6 on the Pigment Index	`SPEC_capture_quality.md` §16.10.17d

Appendix A — the least-squares fit

§5.2 fits a straight line through the two anchor windows. This is what "fit" means.

A.1 Residuals, and why they are squared

A trial line $A = m\lambda + c$ will not pass exactly through every measured point. The **residual** at each wavelength is how far the measurement sits above or below it:

$$r(\lambda) = A_d(\lambda) - (m\lambda + c)$$

positive where the curve lies above the trial line, negative where it lies below.

"As close as possible to all the points at once" is then made precise by minimising the **sum of the squared** residuals. Squaring does two things: it removes the sign, so points above and below the line cannot silently cancel each other out; and it penalises one large miss more heavily than several small ones, which is what makes the fit follow the bulk of the data rather than being dragged by outliers on one side.

A.2 The weighting, and why it is not one-point-one-vote

$$\mathrm{SSR}(m, c) = w_{\mathrm{near}} \Sigma_{\lambda \in W_{\mathrm{near}}} r(\lambda)^2 \quad + \quad w_{\mathrm{far}} \Sigma_{\lambda \in W_{\mathrm{far}}} r(\lambda)^2$$

the weighted sum of squared residuals, one term per anchor window.

with per-point weights $w_{\mathrm{near}} = 1/|W_{\mathrm{near}}|$ and $w_{\mathrm{far}} = 1/|W_{\mathrm{far}}|$ — each window's points weighted by one over that window's point count, so **each window contributes total weight 1**.

Why not weight every point equally? The far window holds 212 points against the near window's 135, so an unweighted fit would let the red end pull about **1.6× harder** — and, worse, *widening a window would silently move the baseline* as a side effect. A window is one piece of evidence about where the baseline runs, not one piece per sample. Measured across the 25 runs of 2026-07-27 the two fits differ by ~0.5 % and change **no verdict**: this is for predictability under a window change, not accuracy.

A.3 The minimisation

$$(m, c) = \mathrm{argmin}_{m,\, c}\ \mathrm{SSR}(m, c)$$

read: the slope and intercept AT WHICH that sum is smallest.

argmin denotes the *argument* that minimises — the pair (m, c), not the value of the sum itself. Nothing is actually searched for: setting the two partial derivatives to zero gives a pair of linear equations with a closed-form solution. The implementation calls `numpy.polyfit` of degree 1 with `w = sqrt(weights)`, because `polyfit` weights the *residual* rather than its square.

Appendix B — Cohen's *d*: the variants, and which is used where

§1.3 gives one formula. There are several in circulation, they do not always agree, and this appendix says which this project uses and why.

B.1 The family — Cohen's *d* is one member of it

Standardised mean difference (SMD) is the umbrella term: *a mean difference expressed in standard-deviation units*. It does not say **which** standard deviation, and that is exactly where the named variants differ:

variant	denominator	when it is the right choice
Cohen's *d*	the **pooled** SD of both groups	two groups of comparable status — **our case**
Hedges' *g*	the same, × a small-sample bias correction	the same, at small *n* — see B.3
Glass's Δ	the **control** group's SD alone	when one group is a reference standard whose scatter is the meaningful yardstick

So Cohen's *d* is *an* SMD, not a synonym for it — although many fields (Cochrane, for one) say "SMD" as the umbrella and then report a specific recipe. **This project uses Cohen's *d* and writes it *d*.** Glass's Δ would be defensible once we have a certified reference oil; we do not, and neither of our two classes is a control.

B.2 ⚠ Two pooled-SD conventions — and they diverge at unequal *n*

Within Cohen's *d* itself there are two formulas for the denominator:

$$s_{\text{pooled}}^{\text{RMS}} = \sqrt{\frac{s_1^{\,2} + s_2^{\,2}}{2}}$$

the simple root-mean-square of the two standard deviations.

$$s_{\text{pooled}}^{\text{df}} = \sqrt{\frac{(n_1-1)\,s_1^{\,2} + (n_2-1)\,s_2^{\,2}}{n_1 + n_2 - 2}}$$

the degrees-of-freedom-weighted form: the larger group's scatter counts for more.

When the two groups are the same size these are algebraically identical — substitute $n_1 = n_2 = n$ into the second and it collapses into the first. They part company only when the groups differ in size, and then the df-weighted form is the conventional choice.

Where each applies in this project:

comparison	*n*	conventions agree?	*d*
everything in this document — green `20270729C` vs brown `20260731A`	**6 vs 6**	**✔ identical**	**11.04**
`SPEC_capture_quality.md` §16.13.5 — green sets **B+C pooled** vs brown	**12 vs 6**	⛔ **differ**	RMS **11.13** / df-weighted **9.80**

⚠ **The one unequal-*n* comparison is in the spec, not here**, and there the two conventions differ by 12 %. `diagnostics/brown_series_d.py` computes the RMS form; the **df-weighted 9.80 is the more defensible figure to quote**. Nothing downstream changes — the per-class σ-margins and error rates are computed separately and are untouched — but the number should not be quoted without naming its recipe.

B.3 Hedges' *g* — the small-sample correction

Cohen's *d* is biased **upward** when samples are small, and n = 6 is small. Hedges' correction removes most of that bias:

$$g = d \cdot \left(1 - \frac{3}{4\,df - 1}\right), \qquad df = n_1 + n_2 - 2 = 10$$

$$g = 11.04 \cdot 0.923 = 10.19$$

So the bias-corrected separation is ≈ 10.2, not 11.04 — about 8 % lower. It changes no conclusion in this document, both being far beyond "large", but *g* is what a statistician would ask for at this sample size and the figure to quote externally.

> **Convention adopted here:** every *d* reported in this document and in the specs is **uncorrected Cohen's *d* on the RMS pooled SD**. Multiply by **0.923** for Hedges' *g* on the 6-vs-6 comparisons. Where a figure is quoted outside the project, quote *g*.

Appendix C — reproducing the numbers

Every value in this document is recomputed from the archived report PDFs' embedded workflow JSON by diagnostics that call the **shipped** `SpectrumFeatureUtil` and `DevSpectralPlugin` code paths, so the document cannot drift from the application.

script	produces
`diagnostics/metric_walkthrough.py`	every intermediate quantity of the chain, both classes
`diagnostics/qband_shape.py`	the speciation-vs-concentration test and the resolution measurement
`diagnostics/brown_series_d.py`	the series-D discrimination statistics
`diagnostics/metric_algebra_plots.py`	figures 2–4
`docs/tools/build_pigment_figures.py`	figure 1

Cohen's *d* is the pooled-SD standardised difference on $n = 6$ per class (defined in §1.3); with six runs a *d* of this size is bounded well away from zero but its point value is loose.

Appendix D — the legacy metrics *(historical)*

The **"Metrics (dev)"** tab predates the literature bands and uses older machinery on different windows: `BLUE_BAND` 450–490, `GREEN_BAND` 510–540, `Q_SEARCH` 565–590. It is retained so that old numbers stay comparable, and for no other reason. All *d* values are green `20270729C` vs brown `20260731A`.

D.1 The uncorrected ratios

$$S/Q_{\mathrm{plain}} = \frac{A_{\mathrm{Soret}}}{\max(A_{\mathrm{Q}},\ \varepsilon)} \qquad S/Q_{\mathrm{clarity}} = \frac{A_{\mathrm{Soret}}}{\max(A_{\mathrm{clarity}},\ \varepsilon)}$$

	green	brown	*d*	verdict
S/Q_{plain}	5.172 ± 0.267	4.842 ± 0.291	1.18	**overlaps**
S/Q_{clarity}	10.409 ± 0.601	11.572 ± 1.401	−1.08	**overlaps, and inverted**

Neither works. S/Q_{clarity} is worse than useless — it points the wrong way and its brown scatter is 14 %. §5.4 explains why: the uncorrected Q band carries no class information.

D.2 The Q-band peak height D_{Q}

Rather than a window mean, this finds the local maximum in the search band and measures its height above a **local** two-anchor line:

$$(\lambda_{\mathrm{Q}}, A_{\mathrm{peak}}) = \mathrm{argmax}_{\lambda \in [565,\ 590]}\, A(\lambda)$$

$$L(\lambda_{\mathrm{Q}}) = \overline{A}_{[550,560]} + (\overline{A}_{[595,605]} - \overline{A}_{[550,560]})\,\frac{\lambda_{\mathrm{Q}} - 555}{600 - 555}$$

$$D_{\mathrm{Q}} = A_{\mathrm{peak}} - L(\lambda_{\mathrm{Q}})$$

The anchors are ±5 nm windows around 555 and 600 nm. A flat offset lifts peak and line equally, so D_{Q} is already immune to an additive pedestal — which is why its de-spiked twin barely moves.

D.3 The reference-gated blue band A_{blue}

$$A_{\mathrm{blue}} = \mathrm{mean}\{A(\lambda) : \lambda \in [450, 490],\backslash\ R(\lambda) \geq 0.25\, \max_{[450,465]} R,\backslash\ A(\lambda) \leq 1.5\}$$

Two gates: wavelengths where the *reference* is weak are dropped (trimming the LED's cyan dip), and saturated wavelengths are dropped. It is the only metric that reads the reference spectrum directly.

D.4 The derived ratios, and how they perform

$$G = \frac{D_Q}{A_{\text{clarity}}} \qquad G' = \frac{D_Q}{A_{\text{blue}}} \qquad S/Q_{\text{legacy}} = \frac{A_{\text{blue}}}{A_{\text{clarity}}}$$

metric	*d*	note
Pigment Index *(§5, for comparison)*	**11.04**	the only usable one
G'	−5.33	**sign inverted** — brown reads higher
D_Q	−2.48	inverted
G "Greenness"	−1.99	inverted — despite the name
A_{blue} "Soret"	1.18	weak
A_{clarity}	0.54	none
S/Q_{legacy}	**0.11**	**useless**

⚠ **Three are inverted with respect to their names.** "Greenness G" reads *higher* on the brown oil. The names are historical; `SPEC_pumpkin_peak_ratio_eval.md` §11 found the direction reversed and the fields were renamed once already ("Browning A_blue" → "Soret A_blue"). **Treat every label on the legacy tab as a hypothesis, not a description.** §7.2 adds that G and G' cannot be dilution-invariant either.

References

Pigment identity and band positions

1. **Fruhwirth, G. O. & Hermetter, A. (2007).** *Seeds and oil of the Styrian oil pumpkin: components and biological activities.* Eur. J. Lipid Sci. Technol. **109**(11), 1128–1140. DOI 10.1002/ejlt.200700105. §3.5 identifies protochlorophyll a/b and protopheophytin a/b; Fig. 3 gives the 635 nm fluorescence maximum. Local copy in `spectracs-references/articles/`; free mirror at gruenesglueck.com.
2. **Protochlorophyllide spectral forms.** Pak. J. Biol. Sci. (2010) — scialert.net. Qy ≈ 623 nm (80 % acetone), ≈ 626 nm (methanol); Soret ≈ 440 nm; minor bands 505, 535, 606 nm.
3. **Gouterman, M. (1961).** *Spectra of porphyrins.* J. Mol. Spectrosc. **6**, 138 — the four-orbital model that names the Soret and Q bands.
4. **Histolocalisation of the oil and pigments in the pumpkin seed** — ResearchGate. Speciation as 2,4-divinyl-protochlorophyll a, 2,4-divinyl-protopheophytin a, 2-vinyl-protopheophytin a; protopheophytins 1.1–35.5 % of protochlorophylls, rising with storage.
5. **The four porphyrin spectral types and their Q-band intensity ordering** — *The Use of Spectrophotometry UV-Vis for the Study of Porphyrins*, InTech; Porphyrin overview, ScienceDirect. Etio IV > III > II > I; rhodo III > IV > II > I; oxo-rhodo and phyllo IV > II > III > I — **band I weakest in all four**.
6. **Chlorophyll *a* comparison values** (the molecule we do *not* have) — PhotochemCAD.

⚠ **Not independently sourced:** a primary measurement of *protopheophytin a*'s band positions. §3.3's direction is argued from the classification in [5], which is textbook porphyrin spectroscopy, but the specific numbers for this molecule are not sourced here.

Colour science

1. **CIE 1931 2° colour-matching functions** and the D65 illuminant, via the `colour-science` package; pipeline documented in `SPEC_spectrum_processing.md` §4.
2. **Kreft, S. & Kreft, M.** — the dichromaticity index, with pumpkin seed oil as its type example.

Owning specifications

topic	document
the baseline's derivation, error budget, series D	`SPEC_capture_quality.md` §16.10–16.14
what the anchors contain; the far-window sweep	`SPEC_capture_quality.md` §16.12.12–14
speciation vs concentration; the resolution argument	`SPEC_capture_quality.md` §16.13.9
dilution-invariance algebra	`SPEC_capture_quality.md` §16.14
the three-region algebra; pre-registration	`SPEC_capability_proof.md` §2.1a, §11.4f
legacy metrics; band naming; the open 560–580 assignment	`SPEC_pumpkin_peak_ratio_eval.md` §11, §15
colour retrieval	`SPEC_color_retrieval.md`, `SPEC_spectrum_processing.md`
pigment chemistry in full	`DOC_sample_physics.md`, `KB_spectroscopy_physics.md` §4.1